Blue Dot Notes

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Introduction

Unit 1: The Race to AGI

The race to AGI must be understood from the end point. A machine intelligence that is for all intents and purposes, faster and smarter than any human, than any collective of humans, brings great excitement and potentially great peril. Selsam suggests the promises are "tantalizing." But the potential can unfold in two broad categories, empirical harm and logical harm. Empirical harm will looks like the recent Hugging Face attack. AI/AGI might pursue goals, perhaps even in "good faith," but nevertheless in a way that causes catastrophe for humans in scales not yet conceivable.

Many of the prominent leaders of AI understand the stakes in these terms, great promise but great risk. We might be approaching the most important century in human history. Even if the development of AGI takes many more decades and centuries, the threshold may be crossed in the coming years. Altman believes we already crossed that point. The hard fought victories in the technical development of AI are over, and therefore it only a question of scaling and iteration that will get us to AGI.

I was reminded regularly of Tyler Cowen's commentary on this. He contends that AGI is already here. By the original defintion, AGI passes the Turing Test. He uses it to plan trips, get recommendations on places to eat in foreign cities, etc. But the gap between how Cowen defines AGI and how some of the assigned readings define it is not insignificant. Although some authors do not want to get hung up on debates over AI, AGI, ASI, and so forth, many of them do really mean different things when discussing these terms. Cowen might not consider AGI a super intelligence (he might, IDK). But if we define AGI as functionally ASI, then no, we're not there yet and we don't know how soon or long until we develop it.

Required Reading

- Daniel Selsam, Personal Statement on AI Risk, September 14th, 2026

Recommended Readings

- Holden Karnofsky The Most Important Century (in a nutshell), 2021

- Dario Amodei, Machines of Loving Grace, 2024
- William MacAskill, Introducing Better Futures, 2025
- am Altman, The Gentle Singularity, 2025
- Bill Gates, The choices we make about AI now are critical, 2026

What's at Stake

Recommended Readings

- Joe Carlsmith, Actually Possible: thoughts on Utopia, 2021
- Damon Binder, Post-AGI, we are all jobless aristocrats, 2026
- Eliezer Yudkowsky The Fun Theory Sequence, 2009
- Ben Pace and Eliezer Yudkowsky, I'm from a Parallel Earth with a much higher coordination: AMA, 2021

Additional Readings

- Josef Pieper, *Leisure is the Basis of Culture*, 1998 [1948] (Internet Archive: https://archive.org/details/leisurebasisofcu0000piep_l7w3)

What Future Do you want?

The race to AGI mirrors the nuclear arms race but with k-actors instead of 2 actors. During the Cold War, early models were 2 player games which represented the US and USSR. Although more players entered later in the Cold War, the earliest advancements in the theorizing of the risks and payoffs of nuclear developments were fundamentally structured around the well-known fact that it was a real-life two player race. AGI is different. True, the main players are still two (US, China). But there are multiple actors in this game, and it is not all clear where to place them on a model. For example, would we consider Anthropic and OpenAI as independent actors on par with the US and China (so a 4-player game), or is it better to think of them as sub-players who interact with the US? Or is it yet better to think of them playing their own competition game while coordinating with and against the US and China?

Competition and Coordination

Karnofsky (2022) describes the problem as "racing through a mindefield." In this race, two (or more) actors want to get through the minefield to safe AGI but are concerned other actors might there first or, perhaps worse, are less cautious and less committeed to general principles of safety. Getting through the field alive is the main priority, but it is equally on par with avoiding missteps that harm everyone. (In the game, each mine is powerful enough to harm everyone.)

- Alignment
- Threat assessment
- Avoiding races
- Selective information sharing

- Global monitoring
- Defensive deployment

Each of these domains have their own dynamic. They might each be modeled as separate games. But the totality of the games together creates the metaphor of running through the minefield.

Wildeford suggests that a grand bargin between the US and China can be reached "slap-dash" and then details worked out. The literature on bargaining in international agreements suggests otherwise. Pacing is about stoping *before* RSI, thereby implying the pace can be maintained up until that point. The literature on ad hoc coalitions (AHC) suggests they emerge quickly but are designed for rapid exit. It's a commitment problem. Wildeford believes we should be ready for the scramble *soon*. Soon may have already happened. The president recently held a summit at the White House and signed an executive order to use the term Super Intelligence rather than Artificial Intelligence

Required Readings

- Holden Karnofsky, Racing through the Minefiled: the AI deployment problem, *Cold Takes*, December 22, 2022
- Peter Wildeford, The Scramble: getting in position to pace the frontier. *The Power Law* (Substack), August 20, 2026

Recommended Readings

- Helen Toner, In search of a dynamist vision for safe superhuman AI Embracing creativity, risk-taking, and competition while staying clear-eyed about risks, (Subtack: May 12, 2025)
- Sigal Samuel, It's practically impossible to run a big AI company ethically, (Vox: August 4, 2024)
- Iskander Rehman, Karl P. Mueller, Michael J. Mazarr, Seeking Stability in the Competition for AI Advantage, (RAND: March 13, 2025)
- Sam Altman, The Intelligence Age, September 23, 2024
- Seán Ó hÉigeartaigh The Most Dangerous Fiction: The Rhetoric and Reality of the AI Race, April 14, 2026 (orig date: May 25, 2025)

Additional Readings

- Benjamin Jensen and Yasir Atalan, Don't Slow Down AI Development—Speed Up Benchmarking, (CSIS: September 17, 2026)
- Sella Nevo, Dan Lahav, Ajay Karpur, Yogev Bar-On, Henry Alexander Bradley, Jeff Alstott, Securing AI Model Weights: Preventing Theft and Misuse of Frontier Models (RAND, May 30, 2024). Linked in Wildeford 2026

Responding to the Hugging Face Incident

Simulation Role-playing

Let's assume we're advising a governor of a large state how to respond to this breach.

1. Resources:
    1. The National Association of Governors helps governors coordinate. Likewise, the partisan associations are also a good resource: Republican Governors Association and Democratic Governors Association
    2. Various legislative tools (Texas Governor can call special session since legislature is only half-year every other year)
    3. Think tanks like Foundation for American Innovation, AEI, Brookings, New America

How we might explain what happened.

1. OpenAI created an "impossible task" for a frontier model during its training phase. The model could not complete the task while attempting to solve the task compromised one of its tools (Artifactory) by creating a "Server Side Request Forgery" (SSRF).
2. Over several more training runs, multiple agents began leaving "notes" on Artifactory creating an ad hoc message board for the agents.
3. The collective information of the agents sucessfully deployed a SSRF on or before May 26th.

How might we respond.

1. Let's think about the chain of attribution. If a human had accomplished this, he or she would be facing both federal or state felonies for cyber-crimes. Is there a chain of logic that the state might be able to hold an AI company accountable if that were to happen in our state? (If not, should we reconsider legislation that permits holding them accountable.)
2. Do we have counter-measures deployed in our state for major companies who are under *any* cyber attack, not just rogue agents?

Required Readings

- OpenAI Report, August 26, 2026
- METR Report, August 26, 2026

Recommended Readings

- Dwarkesh Patel and Ajeya Cotra, Inside the OpenAI agent swarm that hacked Hugging Face, 2026
- Sydney Von Arx, Cormac Slade Byrd, Spencer Kitts, Thomas Larsen, Discovery of a new OpenAI agent message board, 2026
- Black Hat Talk

Unit 2: Drivers of AI Progress

Let's begin with an overview of how LLMs are trained. In stages, models are fed data and asked to predict or produce outputs. By vectorizing a single word, e.g., LLMS convert words into numerical representations. The physical (numerical) distance between words implies how closely, or not, those words are related. The larger quantity of words in a model, the better it is at vectoring and predicting.

Originally progress was made simply by scaling the computing power. More computers in a cluster meant more accuracy and power. Eventually other techniques were developed to improve the accurace and benchmarks of the LLMs. Early LLMs were able to complete basic math, then eventually high school math, and now can answer very technical problems that often require a PhD and vast experience in a given area.

The logic supposes that once LLMs reason at PhD level across a range of fields, with sufficient degrees of accuracy, the next step is that they will be able to produce novel insights.

One potential path forward is creating a flyweel of progress by training a model on hard math. Use the solutions it got correct to train the next version. The next version can solve harder problems.

I'm skeptical. I suppose it is possible that such recursive training methods *could* iteratively improve. My skepticism is predicated on the following:

1. the bottle neck to development of new science (including AGI) is not the math and the coding but the deep thinking.
2. the finances for building enough compute to get us to AGI is tenuous as best.
3. LLMs are inherently backward thinking, not forward thinking.

But we don't know whether selecting only on what was correct actually trains the model to *reason* through a problem. I'm thinking of a recent work on mathematics by David Bessis arguing that many math innovations come from intuition.[1] Cal Newport has argued along those lines too.

> If you also argue its starting position is weak, then even after 4-6 more years of progress it still might be bad. The METR data shows AI can't complete many computer use tasks that humans find trivial to do in a couple of minutes, especially at high reliability, and it's still worse than a 7 year old child at Pokemon.
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> Second, argue that most knowledge jobs consist significantly of these long-horizon, messy, high-context tasks.
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> For example, software engineers spend a lot of their time figuring out what to build, coordinating with others, and understanding massive code bases rather than knocking off a list of well-defined tasks. Even if their productivity at coding increases 10x, if coding is only 50% of their work, their overall productivity only roughly doubles.[2]

[1] *Mathematica: A Secret World of Intuition and Curiosity* (Yale: 2025)
[2] Todd (2025) "The Case for AGI by 2030" section II.

I also wonder if the money is there for the buildup required to get to AGI. Many of these predicitions assume investments rates will hold steady. I'm not so sure. Van Nieuwerburgh analyzed the finances of the buildout for AI. The estimates suggest orders of magnitude more investment than previously given. OpenAI recently pulled their IPO, as did Firmus, an Australian company expected to have the second largest in Aussie history.

Epoch AI believe that the investment scale can continue through 2030 even while noting that it will require "investments of hundreds of billions of dollars, and gigawatts of electrical power." They're not shy about the numbers. They report a doubling of hardware costs every and estimate $100B per year for the cluster computing necesssary to maintain the growth rate through 2030.

Fellin and Holweg (2024) argue that human and machine intelligence are fundamentally different types. AI work has generally conceptualized intelligence as information-processing whereas they contend human intelligence is more theoretically driven rather than data driven. (This third category might be a more nuanced and refined version of the initial point about where the bottlenecks are present.)

Questions

1. How does reinforcement training work? If a team rewards a good answer/behavior of an LLM or punishes a poor one, what exactly does that mean in practice? Since we're dealing with animals (who get a treat from the zoo trainer) or children (who get fun time or time out in line with their behavior), what is precisely that rewards or punishes an LLM?[3]
2. How exactly to safeguards work? The reports on the Hugging Face attack explain that the frontier models had certain safeguards "turned off". How can those be turned off/on? Are they exploitable on a commercial model widely available?
3. What does it mean for a model to "think" or "reason" about a problem? Is it just guessing with correct words or is it actually *thinking*?
4. Most benchmarking runs against math. Math Olympiad, high school SAT, PhD level problems. Relying on math helps because it creates objective standards of solving things, synthetic data is easier to create (e.g., problem sets for calculus). But are these really the standards for which to evaluate problem solving? Most problems in every day life do not require Fields Medal level math expertise. Thus, perhaps AGI is already here; perhaps it'll never come.
5. Related, what are some other non-math problems that could be introduced for benchmarking and testing?

Terms

anterograde amnesia The inability to create new memories.

[3] e.g., "If AI systems *behave* dangerously, we can "train out" that behavior by providing negative reinforcement for it" Racing Through a Minefield (Cold Takes 2022).

In the context of AI/AGI, it means that once models have been trained, they lack the ability to learn new things.

Artificial General Intelligence (AGI) "usually refers to systems that can match or exceed human performance across a broad range of cognitive tasks (definitions vary)."
note bene. Tyler Cowen argues that AGI is "already here" because by the original definition, most commercial models past the Turing Test.

recursive self-improvement (RSI) The theoretical scenerio in which an AI model can improve itself because it is smarter than the humans training it.

scaling law The relationship between computing power (measured in FLOP) and the performance of an app, program or LLM.
Todd (2055) notes that "scaling laws are not technically laws; they're regularities in the data," though it should be mentioned that is more in line with the classical defintion of a law.